

Questions on the manuscript and the talk

1 October 2026

Seven questions, each answered from an artifact in the repository rather than from recollection. Sections 1 and 2 concern one sentence of section 4 of the manuscript; sections 3 and 4 concern one sentence of the speaker notes; section 5 concerns the extremal solutions of [arXiv:2303.12471](#); section 6 whether our solver could construct them; and section 7 the mass-curve secants of the result slide. What was *not* checked is stated at the end of each answer and collected under Provenance.

1 What does 356 mean, and are the couplings equispaced?

The sentence at issue opens section 4, *Results*:

“The dataset contains 356 accepted solutions at 13 couplings from $\alpha = 0$ to $\alpha = 1/2$ in units $r_H = 1$.”

The 356 is a total over couplings, not a count per coupling

It is the number of converged black-hole solutions *summed over all thirteen couplings* — not 356 per coupling, and nothing to do with the number of α values. Each coupling carries one *walk*: a continuation in the spin at that fixed α (section 2 below). The number 356 is the total length of the thirteen walks.

α	states	walk	$TJ^{1/3}$ range
0	43	full	1.12×10^{-5} to 8.74×10^{-2}
0.005	10	narrow	2.44×10^{-3} to 5.10×10^{-2}
0.01	11	narrow	1.21×10^{-3} to 5.14×10^{-2}
0.02	40	full	4.61×10^{-3} to 8.15×10^{-2}
0.035	11	narrow	3.76×10^{-3} to 5.13×10^{-2}
0.05	45	full	4.49×10^{-4} to 7.42×10^{-2}
0.075	16	narrow	4.38×10^{-4} to 4.75×10^{-2}
0.1	45	full	3.84×10^{-4} to 6.48×10^{-2}
0.15	17	narrow	2.57×10^{-4} to 3.90×10^{-2}
0.2	45	full	1.44×10^{-4} to 5.21×10^{-2}
0.3	16	narrow	1.06×10^{-4} to 2.59×10^{-2}
0.4	13	narrow	5.06×10^{-5} to 1.97×10^{-2}
0.5	44	full	3.34×10^{-6} to 3.34×10^{-2}
total	356		

Table 1: States per coupling, read from `results/egb-extremality.json`. The two populations are the two kinds of walk described in section 2.

The counts are bimodal, and the split is exactly the one the next sentence of section 4 describes. The six couplings walked over the *whole* spin range, $\alpha = 0, 0.02, 0.05, 0.1, 0.2, 0.5$,

carry 40–45 states each, 262 in total. The seven that start at $q = 0.6$ and run only to the near-extremal end carry 10–17 each, 94 in total. So

$$356 = \underbrace{262}_{6 \text{ full walks}} + \underbrace{94}_{7 \text{ narrow walks}}.$$

“Accepted” means the state passed the solver’s acceptance gate: the stored rows carry the diagnostics it is judged on (`max_relative_tensor_residual`, `psi_window_spread`, `mass_tail_spread`, and so on). Refused attempts are not in the count.

The couplings are not equispaced

α	0	.005	.01	.02	.035	.05	.075	.1	.15	.2	.3	.4	.5
gap		.005	.005	.01	.015	.015	.025	.025	.05	.05	.1	.1	.1

The spacing is graded, widening by a factor of about 20 from the small- α end to the large- α end. That matches what is being measured: the result is the departure from the first-order value

$$M_{\text{ext}} = \frac{3}{2}\pi^{1/3}J^{2/3} + \pi\alpha + \mathcal{O}(\alpha^2),$$

so the $\alpha \rightarrow 0$ end has to be resolved finely enough to see the slope leave π , while the large- α end only has to track a slowly varying curve.

The grid is also aligned with its two uses. The six full walks sit at 0, 0.02, 0.05, 0.1, 0.2, 0.5, and the seven narrow ones interleave between them. The narrow walks exist only to add resolution to the extremal-slope curve, which is why they skip the low-spin region entirely.

2 What is a “walk”?

A *walk* is one continuation sequence: a chain of solutions at *fixed* α , stepped in the spin, each solve seeded by the previous one. It is the unit the dataset is organised into — thirteen couplings, thirteen walks, 356 states. What follows is read from `src/rotating_bh/extremality.py`.

Where a walk starts

`walk_to_extremality(alpha_gb)` first builds a base solution with `predictor_ladder(start=.60, ...)`: it climbs in the *coupling* from the known $\alpha = 0$ Myers–Perry solution up to the target α in steps of `coupling_step=.005`, holding the spin at $\Omega_H = 0.6$. In units $r_H = 1$ one has $q = r_H\Omega_H = \Omega_H$, which is why the manuscript calls this $q = 0.6$.

Where a walk goes

From that base it runs `walk` twice, in both directions in the spin, with `spin_step=.01` in Ω_H :

- `direction=-1`, down to $\Omega_H = 0$, the static limit;
- `direction=+1`, up until the temperature falls below `floor=1e-6`, the near-extremal end.

The walk is the concatenation `descending + [anchor] + ascending`, sorted in Ω_H . Its length is that coupling’s row in the table above. This is precisely the six-versus-seven split: the couplings given both directions are the 40–45-state full walks, while the seven narrow ones were given only the ascending branch from $q = 0.6$ — the region the $T \rightarrow 0$ extrapolation actually consumes.

Why it is a walk and not a scan

Every step is Newton-solved from the previous solution as its seed. These states cannot be solved independently, because near extremality there is no good initial guess except a neighbour. Three behaviours of the loop follow from that, each recorded in the source as the answer to a way an earlier version failed:

- **Halve, do not stop.** A Newton failure is treated as a seeding accident rather than a physical boundary: the step halves and retries. What ends a walk is the step collapsing below `minimum_step=1e-5`. The docstring puts it plainly — “a wall which looked physical was not”.
- **Roll back on a refused state.** A state that misses the acceptance gate is not kept as the seed. Otherwise every later solve inherits the bad profile and the walk dies far short of where it could reach, which the source records as what the first production run did at the two smallest couplings.
- **Raise the resolution before shortening the step.** Approaching extremality the independent tensor residual grows, and the cure is $N = 64 \rightarrow 96$, not a smaller step. The resolution that accepted each state is stored with it, which is the `resolution` field in the rows.

Why 356 is not simply steps times couplings

A walk steps far more often than it records. States are kept on a schedule: whenever the spin has moved by `spin_record=.02` or the temperature has changed by a factor of `temperature_record=1.3`. Measuring at every accepted step would spend the run in `diagnose`; measuring on a fixed spin grid would crowd the samples where nothing changes. So 356 counts *recorded*, *accepted* states, and the number of Newton solves behind them is substantially larger.

This also explains the temperature coverage in the table of section 1. The $1.3\times$ temperature trigger makes the samples bunch up automatically as T collapses towards extremality — which is where `extremal_state` needs them, since it fits only the `window=12` coldest states.

3 What is the “rotating solver”?

The sentence at issue is in the speaker notes for the cost slide:

“Reproducing all known solutions took about four hours; the first new object, the rotating solver, took five times more ...”

It is the code that finds rotating Einstein–Gauss–Bonnet black holes: `src/rotating_bh/egb_rotating_bvp.py`, built in milestone (*hito*) 4. The ten-minute notes say it at greater length — “the first genuinely new object, the rotating Gauss–Bonnet black hole” — which is what the five-minute line compresses.

What it is

A spectral collocation solver for the radial boundary-value problem of the rotating EGB system. From its docstring:

- Chebyshev–Lobatto nodes; damped Newton with a *complex-step* Jacobian and a line search — the same architecture as the vacuum solver `vacuum_bvp.py`, deliberately mirrored.

- Four compact amplitudes B, F, H, W on a compactified radius, $z = 1/r$ and $x = 1 - z$, so that $x = 0$ is the horizon $r_H = 1$ and $x = 1$ is $r \rightarrow \infty$.
- The horizon row is singular in the generic interior formula — the coefficient matrix of the underlying $[f', b'', h'', w'']$ system degenerates there — so it uses separately derived horizon relations, verified against the exact Myers–Perry closed form.
- At infinity, $B = 1, F = 1, H_x = 0, W_x = 0$.
- At $\alpha_{\text{GB}} = 0$ the interior equations reproduce the vacuum ones exactly.

Everything in the paper rests on it: the walk of section 2 calls into it at every step, and the 356 states are its accepted outputs.

Why it was the expensive block

Hito 4 cost 11.2 hours and 508 M tokens, 31% of the project’s time and 36% of its tokens — the largest single block. The “five times” of the notes is a token ratio, and it is a comparison across different units:

Block	Hours	Tokens
Hitos 0–3, reproducing the known literature	4.4	101,558,821
Hito 4, rotating EGB black hole	11.2	507,954,588

$508/101.6 = 5.0$. Note that the sentence quotes hours for the first block and tokens for the second.

The ledger documents why it was costly:

- **The direct tensor derivation was symbolically intractable.** Christoffel and Riemann for the non-diagonal metric did not terminate. The work moved to the reduced action, with equivalence to the full tensor verified *numerically* over generic profiles rather than symbolically.
- **Five discarded solver routes** before the one that worked: fitting a single horizon parameter; multi-parameter shooting with `least_squares` and `solve_ivp`; `solve_bvp` seeded from Myers–Perry with an exact analytic Jacobian, which went singular with parameters diverging to 10^4 – 10^{22} ; extracting the horizon conditions from the compactified form, where `sp.limit` did not terminate; and translating the raw relations into compact variables, which gave correct but unrecognisable expressions that failed to reproduce even the vacuum form. The route that worked redid the horizon and infinity expansions *directly* in the compact variables.
- **Three self-inflicted errors**, each costed: z and x treated as independent symbols; the numerator extracted before the series, producing a spurious relation; and f'', w'' omitted from the physical jets, which left an order-one tensor residual even on exact Myers–Perry.

4 Is it a different solver from the one of arXiv:1010.0860, and what was it used to reproduce there?

Different in method and independently written. The second half of the question has the more interesting answer: the rotating solver was never successfully checked against their *bulk numbers at finite coupling*. The single comparison that would do that is the one that failed.

The two solvers

	Brihaye, Kleihaus, Kunz, Radu arXiv:1010.0860	This project egb_rotating_bvp.py
Method	COLSYS, a standard off-the-shelf BVP package (their ref. [41]: Ascher, Christiansen & Russell)	Chebyshev–Lobatto spectral collocation, written from scratch
Nonlinear solve	Newton–Raphson with adaptive mesh selection	Damped Newton, complex-step Jacobian, line search
Grid	10^2 – 10^3 mesh points, adaptive	$N = 64$, raised to 96 on a refused state
Quoted accuracy	relative $\sim 10^{-6}$	tolerance 10^{-11} , with per-state residual gates
Equations	their own reduction	independently rederived via the reduced action, equivalence checked numerically

What the two share is a strategy, not code. They also continued from the closed-form solutions, “increasing gradually α or Ω_H ”, which is structurally the walk of section 2 together with the climb in the coupling that seeds it.

What was actually reproduced against them

The eight external contrasts of `results/egb-rotating-external-contrasts.json`, with their sectors, which are the point:

Check	Sector	Result
reduced normalisation pinned	convention	pass
$\omega_H^2(j^2)$ reduced relation	Einstein-limit	pass
each-spin reading of J	convention	pass
critical temperature	Einstein-limit	pass
Wald entropy matches paper	Gauss–Bonnet	pass
$b'(r_H) = 2r_H/(r_H^2 + \alpha)$ at $\Omega_H = 0$	Gauss–Bonnet	pass, $\sim 10^{-14}$
b', f' decrease monotonically in Ω_H	Gauss–Bonnet	pass
published ergosurface radii, their §4.1.1	Gauss–Bonnet	fail

To these the manuscript adds their near-horizon eqs. (4.11) and (4.12), satisfied to 9.17×10^{-14} and 6.66×10^{-16} , which is what pins the coupling conversion $\alpha_{\text{ref.}} = 4\alpha$.

The structure matters more than the tally. The file’s own scope note says the contrasts were chosen among statements “that are closed forms or carry more digits than the section 4.1.1 radii”. Hence:

- the Einstein-limit agreements **do not test the Gauss–Bonnet terms at all**, and the file labels them apart for that reason;
- the two tightest Gauss–Bonnet agreements are against *closed-form* statements — the static horizon-derivative formula at $\Omega_H = 0$, which carries no rotation, and the near-horizon algebra, which is computed by `near_horizon.py` and **does not use the bulk rotating solver**;
- the only check that compares our rotating bulk solutions at finite α against values they obtained *numerically* is the ergosurface radius, which gives 1.102101 against their 1.104 at $\alpha = 1/2$ in our convention. The manuscript states that the discrepancy is unresolved and declines to offer it as evidence of an erratum.

So, stated strictly: the rotating solver reproduces Myers–Perry exactly at $\alpha = 0$, reproduces closed-form EGB statements, and reproduces the near-horizon sector — but its finite- α rotating *bulk* output has one published cross-check, and that one disagrees in the fourth digit.

The manuscript’s containment argument is that the ergosurface radius tests the reconstruction of the metric functions *away from the horizon*, whereas the result depends on the asymptotic charge extraction, the response solve and the zero-temperature extrapolation, none of which uses r_e . That is a reasonable argument and it is made in the discussion rather than buried; whether it is strong enough is a judgement a referee may well press on.

5 Why do the extremal solutions of arXiv:2303.12471 not give M_{ext} , and with it an independent check?

In principle they do, and the manuscript says so: scale invariance makes the extremal branch one curve in their reduced variables, and that curve fixes $M_{\text{ext}}(J, \alpha)$ through $\mu = \mu(0) j^{-2/3}$ and $y = 4x\mu/(3\pi)$ (manuscript/main.tex, introduction). In practice the paper does not publish the branch in a form that can be compared at the precision of our results. Four separate reasons.

1. The near-horizon solution carries no mass

The exact rotating squashed $\text{AdS}_2 \times S^3$ solution (their eqs. (3.18)–(3.19)) gives J , A_H and $S(\text{extremal})$, not M . The mass is an asymptotic charge and the near-horizon geometry does not fix it; this is also why arXiv:1010.0860 obtained only the extremal entropy.

2. The full extremal solutions give M , but no numbers are printed

The directly constructed extremal solutions do give it, $M = \frac{3\pi^2}{4}(a - c_t)$ with c_t read off the large- r tail (their eq. (3.15)). The paper prints no table of c_t , M or J ; the branch appears only as curves in their Fig. 1, “resulting from the extrapolation of around one thousand of data points into the continuum”.

3. The figure does not show the projection we would need

Every quantity in Fig. 1 is normalised by M , and the extremal curve appears only as two separate projections: a_H against $x = c_0\alpha/M$ in the main panel, and a_H against j in the inset (likewise for s and t_H). There is no j against x , which is what fixes $\mu(y)$. Recovering it means matching the two projections point by point through a_H . As far as can be seen at the resolution inspected, the inset’s extremal curve is flat and not monotonic near $j \approx 0.95$ –1, so that matching is ill-conditioned there. That is where our branch lives: $j_{\text{ext}} = [\mu(0)/\mu]^{3/2}$ with μ rising by 31% gives j from 1 down to about 0.67.

4. Our result is a slope, and slopes read off a figure are the weak point

The headline numbers ($\pi \rightarrow 1.07 \rightarrow 1.44$, and the ten checks below 10^{-4}) are values of $\partial M_{\text{ext}}/\partial\alpha|_J$. Differentiating a curve read from a figure, or differencing separately constructed extremal solutions, amplifies the error. The paper also states no accuracy for the equal-spin extremal solutions: the only error estimate it gives ($\sim 10^{-3}$) is in its section 4, for the single-plane solutions. Even a perfect reading would have no error bar to compare against.

What could still be done

- *A coarse check is feasible.* The effects are large — the slope drops by a factor of about three and first order overshoots by $2.3\times$ — so a curve good to the percent level could confirm the sign and the overshoot.
- *Getting the curve.* The PDF’s Fig. 1 contains no raster images, so it is drawn as vector curves; the arXiv source may give the vertices exactly, as was done for `arXiv:1010.0860` (`short-summary/summary.tex`, “Reproducing the published family”).
- *The clean option.* Ask the authors for their extremal data (c_t , c_w , a at each α). The manuscript’s discussion already names a direct construction of the extremal branch as what would remove the extrapolation in temperature.

6 Can our solver find extremal solutions, and would constructing them be an independent check?

Not as it stands; a direct construction would be a worthwhile extension, though not a fully independent one.

Why the present solver cannot reach $T = 0$

The ansatz of `src/rotating_bh/egb_rotating_bvp.py` is built around a non-degenerate horizon: $b = (1 - z^2)B$ has a simple zero at $r_H = 1$, the horizon row uses regularity relations derived for that case (`_egb_rotating_horizon_generated.horizon`), and the inputs are (r_H, ω_H, α) . An extremal horizon is a double zero; the expansion there changes form and these relations no longer apply. That is why the project approaches $T = 0$ from above — the coldest state has $TJ^{1/3} \approx 3 \times 10^{-6}$ — and why the result rests on a fit in temperature.

`near_horizon.py` does solve an extremal problem, but only the near-horizon $\text{AdS}_2 \times S^3$ algebra (v_1, v_2, v_3, k) , not the full solution out to infinity. It therefore gives S_{ext} , not M_{ext} (section 5, point 1).

What the extension would involve

1. *A solver for the degenerate horizon*, for instance in the gauge of `arXiv:2303.12471` (horizon at $r = 0$, $u = r^2 + a^2$): a series expansion at the horizon whose leading data come from `near_horizon.py`, and the same asymptotic charge extraction as now.
2. *Seeds* from the coldest states of the existing walks, which at $TJ^{1/3} \sim 10^{-6}$ should already be very close.
3. *The slope at $T = 0$ directly*: the same implicit-differentiation response solve applied to the extremal family at fixed J , giving $\partial M_{\text{ext}}/\partial \alpha|_J$ in one linear solve, with no limit in temperature and no first-law argument at $T = 0$.

What it would and would not check

- *It removes the weakest link.* The $T \rightarrow 0$ extrapolation is what the error bars describe, and they are the spread over 21 fits, which cannot reveal an error shared by all of them. Comparing $\mu(y)$ and the slope at the 13 couplings with a direct construction would test it properly.
- *It is not fully independent.* It would reuse the same derived field equations and the same mass extraction, so an error in either would carry over. Independence from those needs

the authors’ data (section 5) or a second-order perturbative calculation; the manuscript’s discussion names both, with the direct construction, as the three ways to sharpen the result.

The main technical risk

In higher-derivative theories the corrections near an extremal horizon can scale as non-integer powers of the distance to the horizon, so the solution is not smooth there; [arXiv:2303.07358](#) (in [papers/](#)) concerns this for Kerr. If it happens here, the horizon series needs those powers and the Chebyshev convergence degrades. [arXiv:2303.12471](#) states that it uses a power series at $r = 0$, which suggests the equal-spin case is regular, but this was not checked.

7 Do the mass-curve secants suffer from the finite-difference problem?

Yes. The open squares of the result slide are secants $\Delta\mu/\Delta y$ between neighbouring couplings, and they have the usual finite-difference trade-off; in this dataset it matters at the smallest couplings. The difference from the textbook case is the source of the noise: not machine rounding, but the error of each extrapolated extremal mass.

The two errors

For neighbouring couplings y_1, y_2 , with $\mu = M_{\text{ext}}/J^{2/3}$:

- *Noise.* Each $\mu(y)$ is a $T \rightarrow 0$ extrapolation with an error $\delta\mu$, so the secant carries roughly $(\delta\mu_1 + \delta\mu_2)/\Delta y$, which grows as Δy shrinks.
- *Truncation.* A secant is the slope averaged over the interval; evaluated at the midpoint it misses about $\mu''' \Delta y^2/24$, which shrinks with Δy .

So there is an optimal spacing, and below it the noise term blows up.

What the data show

From `results/egb-extremality.json`; the noise column is $(\delta\mu_1 + \delta\mu_2)/\Delta y$ from the stored `mu_spread`, and for the first interval it reproduces the stored `mass_only_spread` (0.263).

interval in y	Δy	$\delta\mu$ at the ends	secant noise	secant
0–0.005	0.0048	$1.3 \times 10^{-7}, 1.3 \times 10^{-3}$	0.26	3.07
0.005–0.010	0.0051	$1.3 \times 10^{-3}, 7.7 \times 10^{-4}$	0.40	2.91
0.010–0.021	0.0109	$7.7 \times 10^{-4}, 1.6 \times 10^{-3}$	0.22	2.65
0.021–0.038	0.0177	$1.6 \times 10^{-3}, 8.4 \times 10^{-4}$	0.14	2.22
0.038–0.057	0.0185	$8.4 \times 10^{-4}, 5.7 \times 10^{-6}$	0.05	1.82
$y > 0.057$	0.03–0.10	10^{-5} to 10^{-6}	≤ 0.002	1.47...1.39

- *Small y .* The couplings are closely spaced and their extrapolated masses are uncertain at the 10^{-3} level, so the secants are uncertain by 10–40%: the blow-up. That the squares land near the curve there is supported by the sampled values, not by their error estimates.
- *Larger y .* Masses are good to 10^{-5} – 10^{-6} and the spacing is wider, so the noise is negligible; the main error is the interval average, which is why the squares sit slightly off the dots around the minimum.

A rough optimum

Balancing the two errors gives $\Delta y_{\text{opt}} \approx (48 \delta\mu/|\mu''|)^{1/3}$. Near $y \approx 0.03$ the slope samples give $|\mu''|$ of a few hundred; with $\delta\mu \approx 10^{-3}$ that puts Δy_{opt} near 0.05, about ten times the spacing used there, and with $\delta\mu \approx 10^{-5}$ near 0.01. At small coupling the secants are noise-limited, and halving the spacing would make them worse.

Why the main method avoids it

The thermodynamic points difference nothing in α : Ψ comes from one linear solve with the Newton Jacobian at fixed α , so there is no $\Delta\alpha$ to choose (backup slide A2). Its only error is the extrapolation of Ψ itself. That is also largest at small coupling — about 0.09–0.16 at $y = 0.005$ –0.04, against 0.001–0.003 at large y , because those couplings were walked over a narrow spin range and their coldest states are not very cold — but it is still smaller than the secant noise there (0.11 against 0.26 in the first interval). This is the manuscript’s argument for measuring the derivative rather than differencing masses.

Provenance

The per-coupling counts, the α grid and the temperature ranges were read directly from `results/egb-extremality.json` and reproduce the generated macros `\Nstates = 356`, `\Ncouplings = 13`, `\Nmapcouplings = 6`, `\Nnarrowcouplings = 7` and `\Nmaplist` of `manuscript/numbers.tex`. The description of the walk is read from `src/rotating_bh/extremality.py`.

The description of the rotating solver is read from the docstring of `src/rotating_bh/egb_rotating_bvp.py`; the costs, the five discarded routes and the three errors from `prompts/uso-de-tiempo-y-tokens.md`, sections 2 and 3. The method of [arXiv:1010.0860](#) is from its section 4.1.1 and its reference [41], in the copy under `papers/`. The contrast table is read from `results/egb-rotating-external-contrasts.json`; the ergosurface numbers are from the discussion section of `manuscript/main.tex`. Section 5 reads eq. (3.15), section 3.2, eqs. (3.18)–(3.20), Fig. 1 and the error statement of section 4.1 of [arXiv:2303.12471](#), in the copy under `papers/`; the relations $\mu = \mu(0) j^{-2/3}$ and $y = 4x\mu/(3\pi)$ and the 31% growth are from `manuscript/main.tex`. That near-horizon data cannot fix an asymptotic charge is standard background, recalled rather than checked here. Section 6 reads the docstring and residual of `src/rotating_bh/egb_rotating_bvp.py`, the module list of `src/rotating_bh/`, the coldest temperature and the 21 fits from the speaker notes, and the closing paragraph of the discussion of `manuscript/main.tex`. The non-smoothness of extremal horizons in higher-derivative theories is recalled, not read from [arXiv:2303.07358](#). Section 7 reads `mu`, `mu_spread`, `psi_uncertainty` and `y` of each extremal record, and `mass_only` and `mass_only_spread` of `route_comparison`, from `results/egb-extremality.json`.

Not checked: no walk was re-run and no solve was performed; the 356 stored rows were not individually re-tested against the acceptance thresholds, and the external contrasts were read from the stored artifact rather than recomputed. The ratio $508/101.6 = 5.0$ of section 3 is arithmetic on the ledger table; the notes do not show the division. For section 5, nothing was digitised and no vector extraction of Fig. 1 was attempted; the shape of the inset curve near $j = 1$ was judged by eye from a 110 dpi rendering of the page. Section 6 is a proposal: no extremal solve was attempted, the horizon expansion at a double zero was not derived, and whether the equal-spin extremal horizon is smooth was not checked. In section 7 the noise column and the optimal spacing are estimates made here, not results of the project; $|\mu''|$ is read roughly from differences of the slope samples, and no secants were recomputed at other spacings.